

$$F = \frac{q_1 q_2}{4\pi\epsilon_0 \epsilon r^2} \quad \Phi = \int_S \mathbf{B} \cos \alpha \, dS \quad f = \frac{v}{\lambda} \quad W_n = \frac{k(\Delta x)^2}{2} \quad C_v = \frac{i+2}{2} R \quad I = \frac{U}{R} \quad \langle D \rangle = \frac{n_2 - n_1}{\lambda_1 - \lambda_2} \quad \vec{a} = \vec{a}_n + \vec{a}_\tau \quad \langle v \rangle = \frac{\Delta S}{\Delta t} \quad S = S_0 - S_1 \quad V = \text{const}$$

$$\vec{E} = \sum_{i=1}^N \vec{E}_i \quad \psi(x) \quad \frac{1}{\lambda} = R z^2 \left(\frac{1}{m^2} - \frac{1}{n^2} \right) \quad h = 6.63 \cdot 10^{-34} \text{ Дж} \cdot \text{с} \quad A = A_0 e^{-\beta t} \quad A = p(V_2 - V_1) \quad A = \frac{m}{\mu} k T h \frac{v}{v_0}$$

$$v = \frac{1}{T} \quad R = \sigma T^4 \quad T = \frac{2\pi}{\omega} \quad x = \beta T \quad \Psi_n = \sqrt{\frac{2}{l}} \sin \frac{n\pi x}{l} \quad \omega = \sqrt{\omega_0^2 - \beta^2} \quad v_k = \frac{A}{h} \quad Q = \Delta U + A \quad c = \frac{dQ}{m dt} \quad C = c \mu \quad S_2 - S_1 = \int_1^2 \frac{dQ}{T}$$

$$x = A \cos(\omega t + \alpha) \quad \omega = 2\pi\nu \quad \Phi = BS \cos \alpha \quad E = mc^2 \quad h\nu = A + \frac{mv_{\text{max}}^2}{2} \quad \Delta m > 0 \quad \Delta m < 0 \quad C = c \mu \quad m_0 = - \quad \langle \lambda \rangle = (\sqrt{2\pi d^2 n})^{-1}$$

$$\sigma = 5.67 \cdot 10^{-8} \frac{\text{Вт}}{\text{м}^2 \cdot \text{К}^4} \quad W = |\Psi|^2 \quad R = \alpha \sigma T^4 \quad x = A_0 e^{\beta_1} \cos(\omega t + \alpha) \quad p = \frac{mv}{\sqrt{1 - \frac{v^2}{c^2}}} \quad E = h\nu = h \frac{c}{\lambda}$$

$$\lambda_m = \frac{b}{T} \quad b = 2.9 \cdot 10^{-3} \text{ м} \cdot \text{К} \quad V_i = \frac{1}{T} \quad \omega = 2\pi\nu \quad T = \frac{2\pi}{\sqrt{\omega^2 - \beta^2}} \quad \beta = \frac{v}{c} \quad \Delta N = N \frac{4}{\sqrt{\pi}} \tilde{e}^{\frac{1}{2}} \tilde{u}^{\frac{1}{2}} \tilde{v}^{\frac{1}{2}} \quad R = \frac{W}{t \cdot S} \quad p = \frac{2}{3} n \langle \epsilon \rangle = \frac{N}{V} \quad \rho = \frac{W}{t S c} = \frac{1}{c} \quad u = \frac{v}{\sqrt{6}}$$

$$\varphi = \arctg \frac{A_1 \sin \alpha_1 + A_2 \sin \alpha_2}{A_1 \cos \alpha_1 + A_2 \cos \alpha_2} \quad \lambda = vT \quad k = \frac{2\pi}{\lambda} \quad p = \frac{1}{c} \sqrt{W_k(W_k + 2E_0)} \quad \Delta m = Z m_p + N m_n - m \quad \langle z \rangle = \sqrt{2\pi} d^2 n \langle v \rangle$$

$$\Delta f = \pm m \lambda_0, \quad m = 0, 1, 2, \dots \quad W = \frac{1}{2} m A^2 \omega^2 \quad \xi = A \cos(\omega t - kx) \quad \lambda \quad A_f = \frac{f_0}{2\beta \sqrt{\omega_0^2 - \beta^2}} \quad k = 1.38 \cdot 10^{-23} \quad \rho = \vec{p}_1 + \vec{p}_2 + \dots + \vec{p}_n \quad \langle \epsilon \rangle = \frac{3}{2} kT$$

$$M = F \cdot l \quad \Delta \varphi = \frac{2\pi}{\lambda} \Delta x \quad p = nkT \quad E_n = \frac{h^2}{8m l^2} n^2 \quad t_0 = \frac{t_0}{m} \quad \lambda = \frac{h}{p} \quad \varphi = \frac{W}{q_0} \quad \eta = \frac{1}{3} \rho \langle v \rangle \langle \lambda \rangle \quad U = \frac{1}{2} \frac{m}{A} R T \quad \frac{pV}{T} = \frac{m}{\mu} R = 8.31 \quad v = \frac{N}{N_A} \cdot \frac{m}{\mu} \quad \sigma = en(u_n + u_p) \quad f(v) = 4\pi \left[\frac{2\pi m k T}{m_0} \right]^{3/2} \frac{1}{v^2} e^{-\frac{m_0 v^2}{2kT}} \quad \Delta u = \frac{\Delta v}{\sqrt{6}} \quad A = I \Delta \Phi \quad G_2 = \frac{5}{2} \hbar \omega (n+2) \quad \lambda_k = \frac{hc}{A} \quad \vec{E} = \frac{\vec{F}}{q_0}$$

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Syllabus

Part 1 – Wave Mechanics

Wave theory of matter; Schrödinger equation; interpreting the wavefunction; operators and expectation values.

Part 2 – Quantum Mechanics in 1D

Solutions of the time-independent Schrödinger equation: potential barriers; quantum tunnelling; square well potentials; simple harmonic oscillator potential.

Part 3 – Principles of Quantum Mechanics

Time dependence; superposition and measurement; simultaneous eigenfunctions; commutation and uncertainty.

Part 4 – Quantum Mechanics in 3D

Degeneracy; angular momentum; central potentials; single-electron atoms (hydrogen); single-electron atom wavefunctions; Stern-Gerlach experiment; intrinsic spin; Pauli exclusion principle; total angular momentum; atomic structure.

Other practicalities

Lectures:

Mostly handwritten using the visualiser in real time.

Some supplementary notes and figures from time-to-time.

2 lectures per week (see details on your online timetable)

Online resources (Canvas):

Panopto recordings to appear each week.

Each week I will add a scanned copy of my handwritten lecture notes to Canvas.

Non-assessed problem sheets will also be there, but solutions will appear later.

Typed lecture notes for entire course already available there.

Recommended textbook(s)

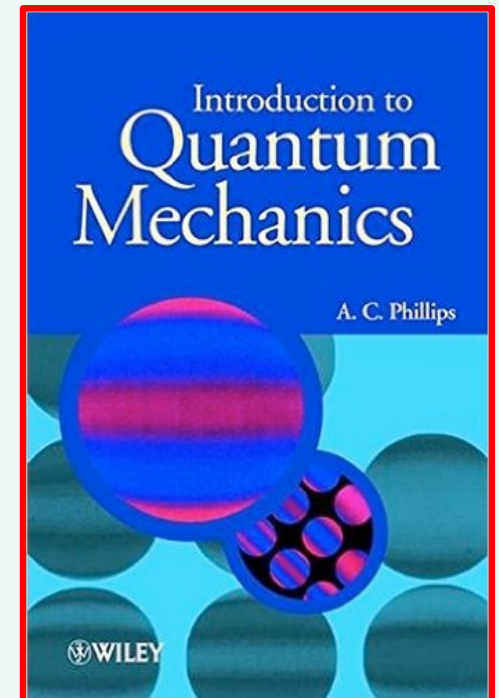
1. Introduction to Quantum Mechanics, A.C. Phillips

2. You may also consult:

Quantum Physics of atoms, molecules, solids, nuclei and particles,

Eisberg and Resnick, 2nd edition

**The right level
Will mostly match course notation**



A few words of introduction

Pretty much everyone who thinks about the world view embodied in QM finds it uncomfortable and highly counterintuitive.

The philosophical interpretation of QM is a very lively subject area which could easily fill many lecture courses and has a vast associated literature. Many of the founders of QM were also influenced by the contemporary philosophy of the early 20th century.

However, most were still unhappy with what they had discovered

‘Anyone who is not shocked by quantum theory has not understood it.’ **Niels Bohr**

‘I think I can safely say that nobody understands quantum mechanics.’ **Richard Feynman**

... and some were never reconciled to it

‘I am, in fact, firmly convinced that the essentially statistical character of contemporary quantum theory is solely to be ascribed to the fact that this (theory) operates with an incomplete description of physical reality’

Albert Einstein

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"I still don't understand quantum theory."

Y1 QM recap

In Y1 QM we saw several phenomena that cannot be explained by “classical physics”, i.e. Newton and Maxwell equations:

black-body radiation spectrum

photo-electric effect

Compton scattering

electron diffraction

Classical approximation breaks down at small scale, of atoms and nuclei.

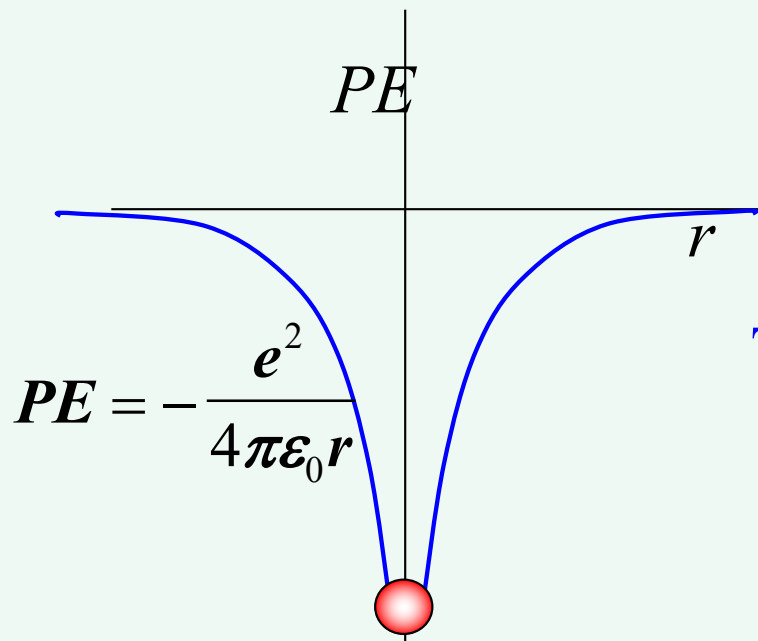
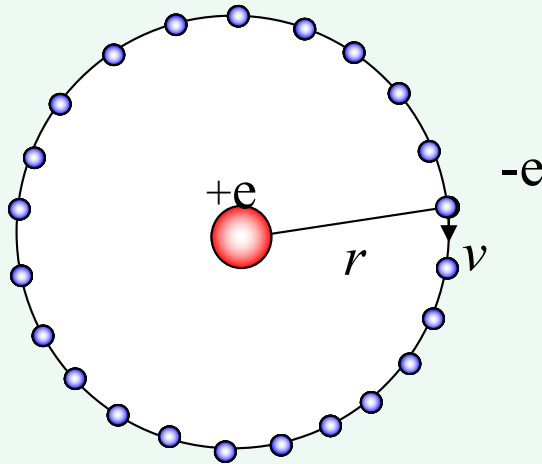
In QM2, we will approach QM more quantitatively, studying first simplest systems (non-relativistic) then move to more realistic systems such as atoms.

We will encounter some non-classical and counter-intuitive behaviour.



The Hydrogen Atom

[Bohr's approach]



Equate
forces:

$$\frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$$

$$mv^2 = \frac{e^2}{4\pi\epsilon_0 r}$$

Potential Energy:

$$PE = -\frac{e^2}{4\pi\epsilon_0 r}$$

Total Energy = $KE + PE$

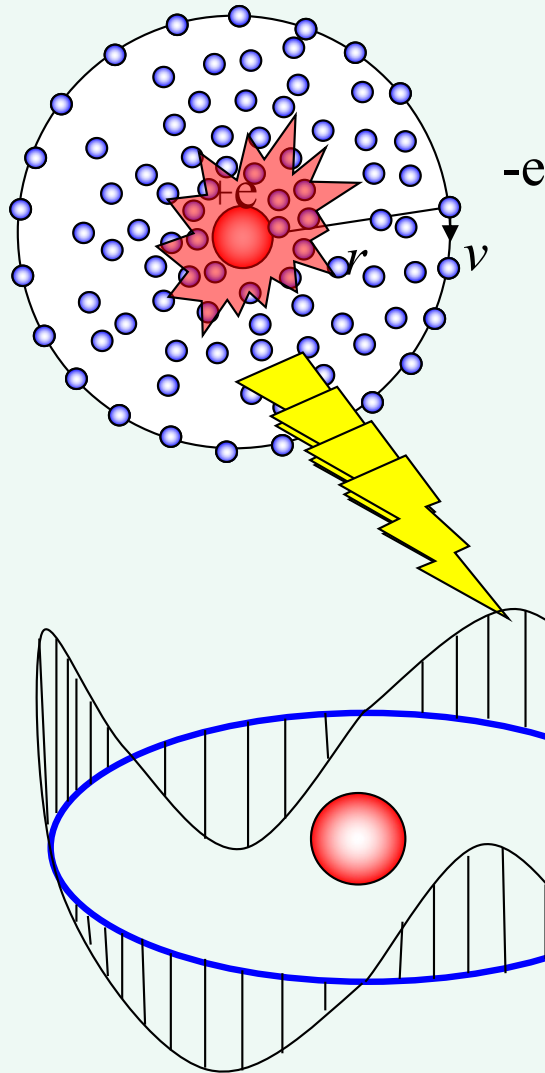
$$= \frac{1}{2} mv^2 - \frac{e^2}{4\pi\epsilon_0 r} = -\frac{1}{2} \frac{e^2}{4\pi\epsilon_0 r}$$

But! When a charged particle accelerates, loses energy.

Quantized orbits:

-Wave particle duality -

De Broglie hypothesis

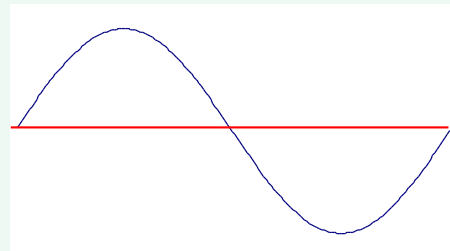


$$p = \frac{h}{\lambda}$$

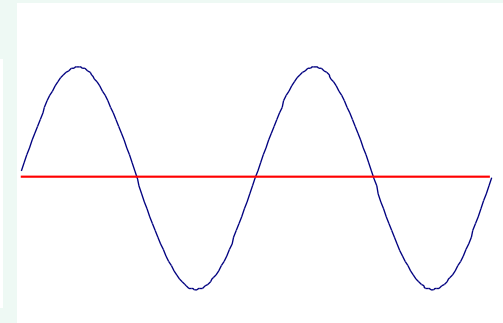
$$\lambda = 2\pi r$$

$$n\lambda = 2\pi r$$

To fit integer number of wavelengths into orbit

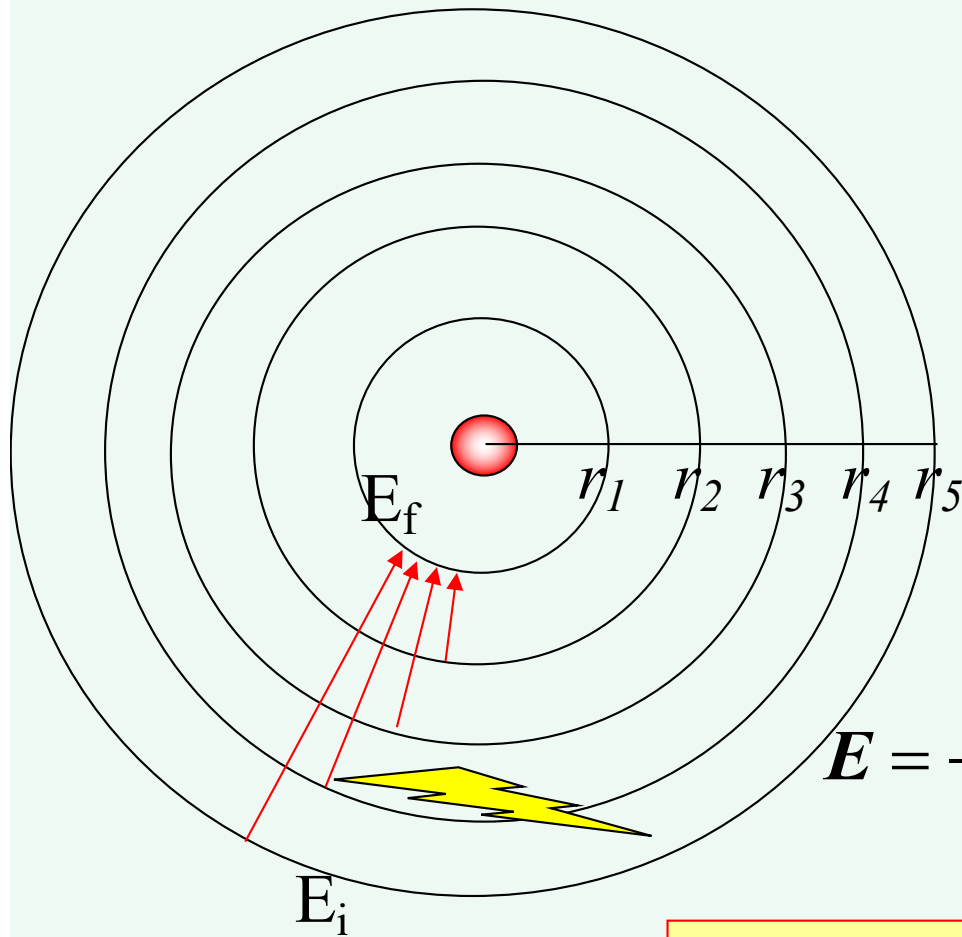


$$p = mv = \frac{nh}{2\pi r}$$



$$mvr = \frac{h}{2\pi} n$$

Quantum Jumps



$$mvr = \frac{h}{2\pi} n \quad mv^2 = \frac{e^2}{4\pi\epsilon_0 r}$$

Eliminate v

$$r_n = \frac{4\pi\epsilon_0}{me^2} \frac{h^2}{2\pi^2} n^2$$

$$E = -\frac{1}{2} \frac{e^2}{4\pi\epsilon_0 r} \quad E_n = -\frac{e^4 m (2\pi)^2}{2(4\pi\epsilon_0)^2 h^2} \frac{1}{n^2}$$

$$E_i - E_f = h\nu \quad \nu = c / \lambda$$

$$\frac{1}{\lambda} = 1.0969 \times 10^7 \left(\frac{1}{2^2} - \frac{1}{n^2} \right)$$

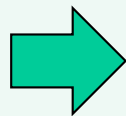
Basic concepts from QM1

- 1) Photon and electrons etc. can behave either as “particles” (e.g. photoelectric effect) or as waves (e.g. diffraction), depending on the type of measurement made
- 2) The “waves” can interfere and superimpose
- 3) Energy of a quantum depends on frequency
- 4) Momentum of a quantum is inversely proportional to wavelength
- 5) These properties are connected through Planck’s constant

$$E = h\nu$$

$$p = h/\lambda$$

$$h = 6.626 \times 10^{-34} \text{ Js}$$



Wave theory of matter

Travelling Wave

$$\Psi(x, t) = A \sin(kx - \omega t)$$

$$t = 0$$

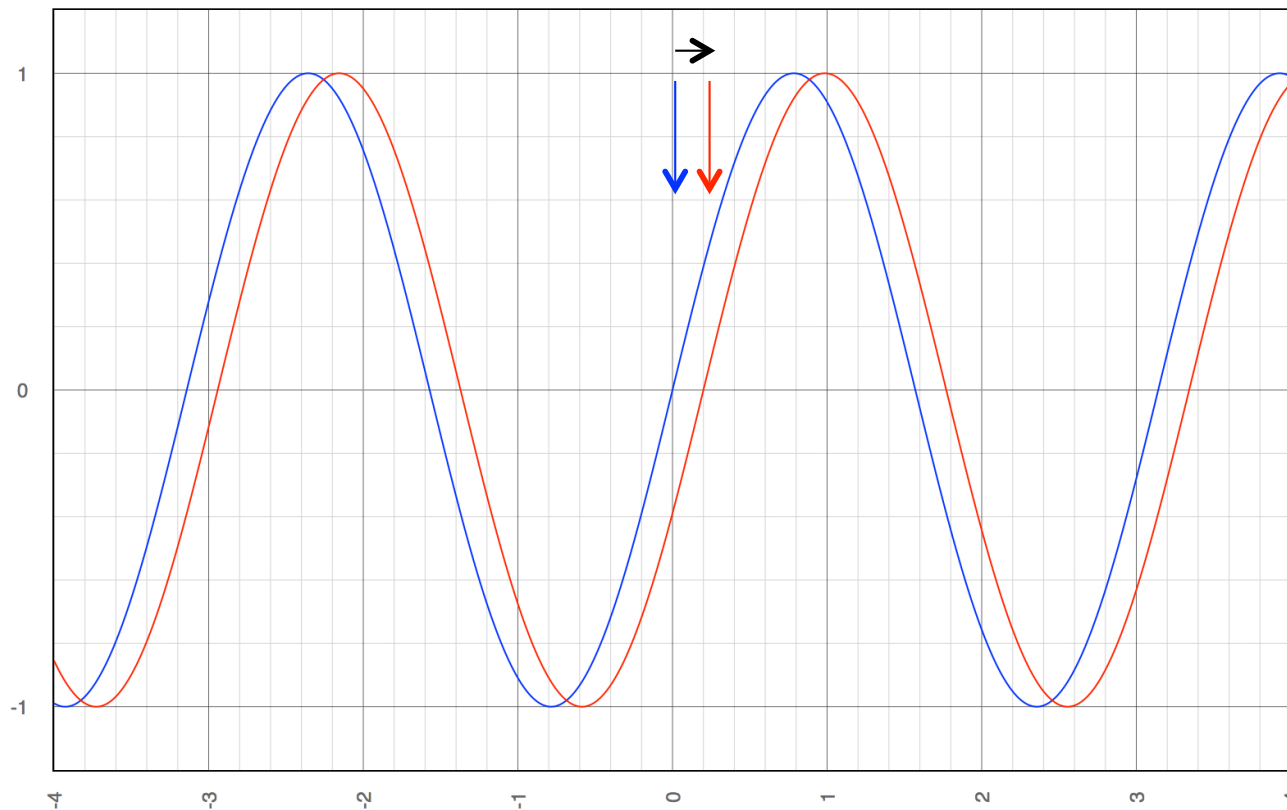
$$\Psi(x, t) = A \sin(kx)$$

Node at $x = 0$

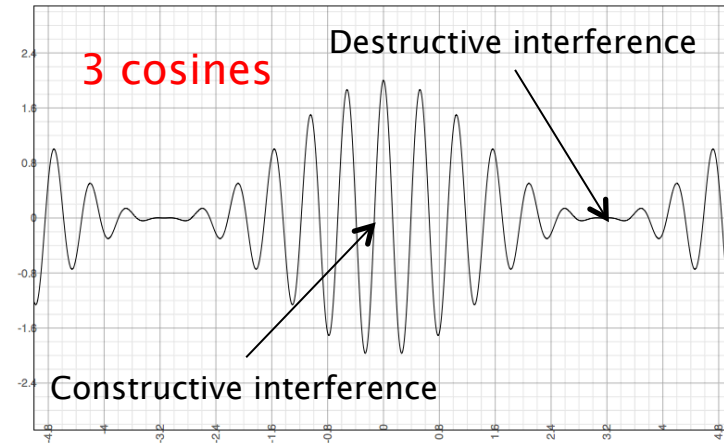
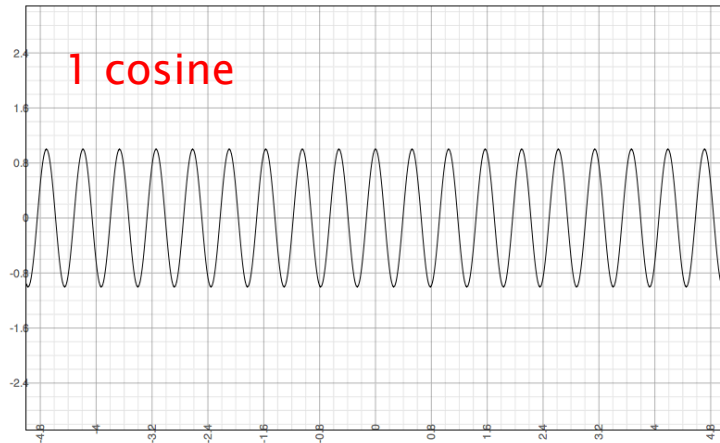
$$t = t'$$

$$\Psi(x, t) = A \sin(kx - \omega t')$$

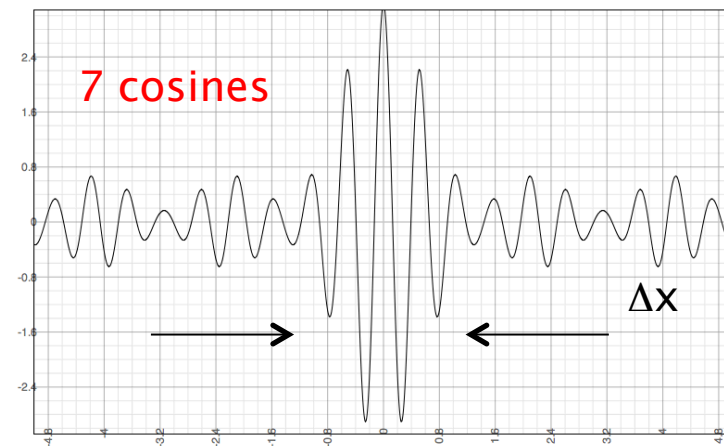
Node at $x = \omega t' / k$



Creating a wavepacket



$$y = \cos 12x + \left(\frac{1}{2}\right)(\cos(11x) + \cos(13x)) + \left(\frac{1}{3}\right)(\cos(10x) + \cos(14x)) + \left(\frac{1}{4}\right)(\cos(9x) + \cos(15x))$$



Summary

(amplitude)² of wave function is proportional to the probability of photons being detected per unit time

Wave theory on matter particles gives probabilities of particles being measured (detected) within a given range (per unit time)

Heisenberg's uncertainty principle expresses intrinsic quantum limit to product of position and momentum uncertainties as a consequence of the wave picture

Next lecture: **Schrödinger equation**